**ECE 329** **Homework 2** **Due: Mon, Sep. 29, 2025, 11:59 PM**

Totaling 70pts.

1. (10pts) Gauss's law for **displacement** $\mathbf{D} = \varepsilon_0 \mathbf{E}$ states that $\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_V \rho dV$ over any closed surface $S$ enclosing a volume $V$ in which **electric charge density** is specified by $\rho(x, y, z)$ C/m$^3$.

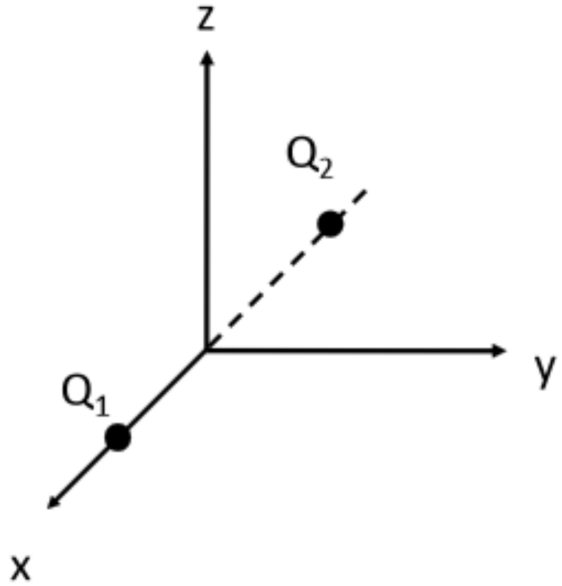
   a) (3pts) Applying dimensional analysis to Gauss' law $\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_V \rho dV$ show that displacement $\mathbf{D}$ is measured in units of C/m$^2$ — carefully describe your reasoning in reaching this conclusion.

   b) (2pts) What is the *displacement flux* $\oint_S \mathbf{D} \cdot d\mathbf{S}$ over the surface of a cube of volume $V = L^3$ centered about the origin, if $\rho(x, y, z) = 3$ C/m$^3$ within $V$ and $L = 1$ m?

   c) (3pts) Repeat (b) for $\rho(x, y, z) = x^2 + y^2 + z^2$ C/m$^3$. **Hint:** you need to compute a 3D integral of $x^2 + y^2 + z^2$ (using $dxdydz$) over the box $V$.

   d) (2pts) What is the displacement flux in part (c) for any one of the square surfaces of volume $V$? **Hint:** take advantage of the symmetry inherent in this problem.

2. (10pts) Two unknown charges, $Q_1$ and $Q_2$ are located at $(x, y, z)$ =(1, 0, 0) and (-1, 0, 0), respectively, as shown below. The displacement flux $\int_{yz-plane} \mathbf{D} \cdot \hat{x} dydz$ through the entire $yz$-plane (i.e., the $x = 0$ plane) in the $+\hat{x}$ direction is 4 C. The flux through the plane $y = 1$ in the $+\hat{y}$ direction is -2 C. Determine $Q_1$ and $Q_2$ after writing a pair of algebraic equations relating the above displacement fluxes to $Q_1$ and $Q_2$.

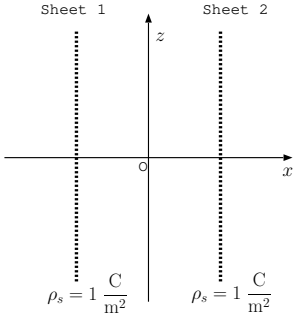



**Hint:** What is the contribution of $Q_1$ to the flux $\int_{yz-plane} \mathbf{D} \cdot \hat{x} dydz$? See Example in Lecture 3.

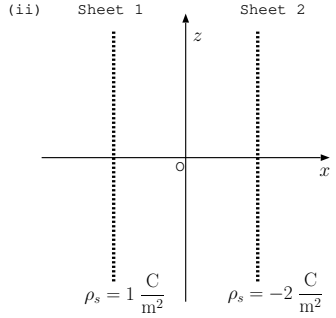
3. (10pts) An infinite charge sheet with uniform charge density $\rho_s \ \frac{\mathrm{C}}{\mathrm{m}^2}$ produces electrostatic fields $\mathbf{E}$ with $\frac{\rho_s}{2\epsilon_o}$ magnitude pointing away from the sheet on both sides. Also, superposition can be used to calculate the fields of composite charge distributions.

Given the above facts, what is the **displacement vector** $\mathbf{D} = \epsilon_o \mathbf{E}$ at origin O in the following diagrams where a pair of charge sheets with specified $\rho_s$ values (defined on $x$ =const. surfaces, with the $y$-axis pointing into the paper) are shown?

(i)




(ii)



4. (10pts) Consider a three-slab geometry of three identical slabs of equal widths $W$ in $x$-direction, and infinite extent in $y$ and $z$ directions. In slab 1 extending over $-2W < x < -W$, charge density $\rho$ is $\rho_o$, where $\rho_o$ measured in C/m$^3$ units is some positive number. In slab 2 extending over $-W < x < 0$ the charge density is $2\rho_o$. Finally, in slab 3 extending over $0 < x < W$ the charge density is $-3\rho_o$. Charge density $\rho$ is zero everywhere else. Determine and sketch $E_x(x)$ over the region $-3W < x < 3W$. Label both axes of your plot carefully, marking the field value at each break point.

   **Hint:** use superposition of shifted and scaled versions of the static field configuration of a single charged slab.

5. (10pts) **Curl** and **divergence** exercises:

   a) On a 25-point graph consisting of $x$ and $y$ coordinates having the integer values $\{-2, -1, 0, 1, 2\}$ sketch the vector field $\mathbf{F} = x\,\hat{x} + y\,\hat{y}$ and find $\nabla \times \mathbf{F}$ (curl of $\mathbf{F}$) and $\nabla \cdot \mathbf{F}$ (divergence of $\mathbf{F}$).

   b) Repeat (a) for $\mathbf{F} = y\,\hat{x} - x\,\hat{y}$.

   c) Based on above results choose the correct answer in the statements below:

      i. $\nabla \times \mathbf{F} \neq 0$ implies the field strength varies (**along** or **across**) the direction of the field.

      ii. $\nabla \cdot \mathbf{F} \neq 0$ implies the field strength varies (**along** or **across**) the direction of the field.

6. (10pts) Use the differential form of Gauss's law to determine the static charge density $\rho(x, y, z)$ in a region of space if:

   a) (5pts) The static displacement field in the region is specified as $\mathbf{D}(x, y, z) = -6\text{sgn}(z)\hat{z}$ C/m$^2$. Describe the nature of the charge density "in words" — do we find a volumetric distribution of charge, or a surface charge, or a line charge? **Hint:** how do we express the derivative of a sgn function?

   b) (5pts) The static displacement field in the region is specified as

$$\mathbf{D}(x, y, z) = \begin{cases} -\hat{z}16\,\text{C/m}^2, & z < -4\,\text{m}, \\ \hat{z}\,4z\,\text{C/m}^2, & |z| < 4\,\text{m}, \\ \hat{z}16\,\text{C/m}^2, & z > 4\,\text{m}. \end{cases}$$

7. (10pts) Coulomb's field of a charge $Q$ stationed at the origin of a right-handed Cartesian coordinate system can be expressed as

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_o r^2}\hat{r} = \frac{Q}{4\pi\epsilon_o r^2}\frac{(x, y, z)}{r},$$

where $r^2 \equiv x^2 + y^2 + z^2$ and $r \geq 0$. Verify that $\nabla \times \mathbf{E} = 0$ by showing that when $\nabla \times \mathbf{E}$ is expanded as usual, all of its components cancel out exactly.